

■ Subject: STEM

■ Grade: 6

■ Duration: 45 min

SECTION 1 • Why Does Our Planet Need Help?

Climate change is making Earth warmer and changing weather patterns worldwide. Earth's temperature has already risen by about **1.1°C** since 1880 — and the pace is accelerating. This warming causes glaciers to melt, sea levels to rise, and extreme weather events to become more frequent.

Our planet needs extra help because natural processes alone cannot keep up with the rapid changes happening to our climate.

■ ■ Greenhouse gases trap heat, causing global warming

■ Nature alone can't remove carbon dioxide fast enough

■ We need new ways to protect plants, animals, and people

■ Climate engineering helps buy time while we reduce pollution

SECTION 2 • What Is Climate Engineering?

Climate engineering (also called **geoengineering**) means using science and technology to intentionally alter Earth's climate system in order to slow down or reverse the effects of climate change.

■ Two Main Types

■ Solar Radiation Management (SRM)

Reflects sunlight back into space to cool Earth down. Uses mirrors, clouds, or white surfaces to bounce heat away.

Examples:

- Giant space mirrors at Lagrange Point 1
- Stratospheric aerosol injection (sulphate particles)
- Marine cloud brightening (sea-salt spraying)
- Urban surface whitening (rooftops, roads)

■ Carbon Dioxide Removal (CDR)

Takes CO₂ out of the air using trees, machines, or by storing it underground safely.

Examples:

- Tree planting / reforestation
- Direct Air Capture (DAC) machines
- Ocean alkalinity enhancement
- Bioenergy with carbon capture (BECCS)

SECTION 3 • Pros & Cons of Climate Engineering

■ Advantages

- Can produce quick results to cool Earth
- Helps protect fragile ecosystems and coral reefs
- Buys time while we transition to clean energy
- Drives innovation and new green technologies
- Some methods (like tree planting) have co-benefits

■ Risks & Challenges

- May alter rainfall and weather patterns globally
- Could reduce pressure to cut emissions (moral hazard)
- Some methods are very expensive and unproven
- Difficult to stop once started (termination shock)
- Raises ethical questions about who decides

SECTION 4 • Climate Engineering Through History

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■ First Ideas

Scientists first imagined ways to cool Earth by reflecting sunlight. These were just theoretical concepts on paper — no technology existed yet.

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■ Research Begins

Researchers started testing carbon capture and cloud brightening in controlled laboratories around the world. Serious academic interest grew.

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■ Pilot Experiments

Small-scale field tests began. Iceland's CarbFix project started mineralising CO₂ underground. Harvard scientists proposed Stratospheric Controlled Perturbation Experiments.

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■ Real Projects Running

Real climate engineering projects are now operational. Trees are being planted at scale, DAC machines are running in Iceland (Orca, Mammoth plants), and solar reflector research continues.

SECTION 5 • Real-World Climate Engineering Projects



CarbFix — Iceland

CO₂ from geothermal plants is dissolved in water and injected into basalt rock, where it turns to stone within 2 years.



Mammoth DAC — Iceland

Climeworks' largest Direct Air Capture plant, opening 2024, removes 36,000 tonnes of CO₂ per year directly from the atmosphere.



Great Green Wall — Africa

An 8,000 km belt of trees across the Sahel region of Africa to combat desertification and sequester millions of tonnes of carbon.



MCBP — Australia

The Marine Cloud Brightening Project on the Great Barrier Reef sprays sea salt to brighten clouds and protect corals from heat bleaching.

■ Fun Fact!

"Science gives us superpowers to protect our planet! Every day, scientists around the world are inventing amazing new ways to help Earth stay cool and healthy."

■ CLASSROOM ACTIVITIES

Activity 1 • Solar Radiation Experiment ■

In this experiment you will compare how different surfaces reflect sunlight.

You will need:

- 3 small boxes or cups
- White paper
- Black paper
- Aluminium foil
- Thermometer (×3)
- Sunny spot or lamp

Steps:

1. Cover Box 1 with **black paper**, Box 2 with **white paper**, Box 3 with **aluminium foil**.
2. Place a thermometer inside each box.
3. Put all three boxes in direct sunlight (or under a lamp) for **15 minutes**.
4. Record the temperature inside each box every 5 minutes.
5. Compare results. Which surface stayed coolest? Which warmed most?

Results Table

Surface	Start Temp (°C)	After 5 min	After 10 min	After 15 min
Black paper				
White paper				
Aluminium foil				

Conclusion: Which material best represents a Solar Radiation Management approach? Why?

Activity 2 • Build a Carbon Capture Machine ■

Put the 6 components of a Direct Air Capture machine in the correct order. Draw arrows connecting each stage and write what it does.

1	■ Air Intake Fan	Draws in air from the atmosphere
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2	■ CO ₂ Filter Sorbent	Binds CO ₂ molecules from airflow
3	■ Regeneration Heater	Heat releases captured CO ₂ (90–120 °C)
4	■■ Gas Separator	Concentrates pure CO ₂ stream
5	■ CO ₂ Compressor	Pressurises CO ₂ to ~150 bar
6	■ Underground Storage	Injects CO ₂ into geological formations

Challenge: How many tonnes of CO₂ does a large DAC plant capture per year? Research and write your answer below.

Activity 3 • Ocean Alkalinity Lab ■

Ocean alkalinity enhancement adds minerals to the ocean to help it absorb more CO₂. Read each scenario and tick the BEST outcome.

Scenario 1: A scientist adds a LOW dose of olivine to a test tank (pH 7.9 → 8.0).

☒ Corals begin to recover slightly

☐ Ocean becomes too alkaline — fish die

☐ No effect whatsoever

Scenario 2: A scientist adds a HIGH dose of quicklime to the ocean (pH 7.9 → 8.6).

☐ Coral reefs flourish and grow faster

☐ The pH is now in the ideal range

☒ Too alkaline — harmful to marine life

Scenario 3: A scientist adds a MEDIUM dose of limestone (CaCO₃), pH reaches 8.1.

☒ Ideal pH for coral growth and CO₂ absorption

☐ The ocean becomes dangerously acidic

☐ Limestone has no effect on pH

■ BONUS QUIZ — Test Your Knowledge!

Q1: What are the two main types of climate engineering?

Answer:

Solar Radiation Management (SRM) and Carbon Dioxide Removal (CDR)

Q2: Name one real-world carbon capture project and where it is located.

Answer:

Mammoth / Orca by Climeworks — Iceland

Q3: How does Solar Radiation Management help cool the Earth?

Answer:

It reflects incoming sunlight back into space, reducing the amount of solar energy absorbed by Earth's surface.

Q4: What is one risk of using stratospheric aerosol injection?

Answer:

It could alter rainfall patterns and monsoons; it must be continuously maintained; stopping it suddenly causes rapid warming.

Q5: Why is it important to also reduce emissions even when using climate engineering?

Answer:

Climate engineering only treats symptoms, not the cause. Without cutting emissions, CO₂ levels keep rising and the planet stays in danger.

■ What Can YOU Do to Help?



Learn — Read about climate change and share what you discover with friends and family.



Plant — Join a tree-planting project in your community or school garden.



Save — Save energy at home — turn off lights, unplug devices, use less hot water.



Speak — Talk to your teachers, parents, and local leaders about climate solutions.

■ The Future of Climate Engineering

Scientists around the world are working on exciting new ideas every day to protect our planet! With curiosity and STEM learning, we can all help make Earth safe and healthy for future generations. **Your ideas might change the world someday!**